Laxmi Charitable Trust's
Sheth L.U.J. College of Arts & Sir M.V. College of Science & Commerce
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Department of Information Technology (Bsc.IT Semester III)
Data structure

Project- Parking Management System

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Code :

```
class Node:
    def __init__(self, vehicle):
        self.vehicle = vehicle
        self.next = None

head = None

removed = []
vehicle_data = {}

def park_vehicle():
    global head
    vehicle = input("Enter vehicle number: ")
    new_node = Node(vehicle)
    if head is None:
        head = new_node
    else:
        temp = head
        while temp.next:
            temp = temp.next
        temp.next = new_node
    vehicle_data[vehicle] = "Parked"
    print("Vehicle parked successfully.")

def display_vehicles():
    global head
    print("\nParked Vehicles:")
```

```
    if head is None:
        print("No vehicles.")
    else:
        temp = head
        while temp:
            print(temp.vehicle)
            temp = temp.next

def search_vehicle():
    vehicle = input("Enter vehicle number to search: ")
    if vehicle in vehicle_data:
        temp = head
        position = 1

        while temp:
            if temp.vehicle == vehicle:
                print("Vehicle found at position:", position)
                return
            temp = temp.next
            position += 1
    else:
        print("Vehicle not found.")

def remove_vehicle():
    global head
    vehicle = input("Enter vehicle number to remove: ")
    if head is None:
        print("Parking is empty.")
        return
    temp = head
    previous = None

    while temp:
        if temp.vehicle == vehicle:
            if previous is None:
                head = temp.next
            else:
                previous.next = temp.next
            removed.append(vehicle)
            del vehicle_data[vehicle]
            print("Vehicle removed successfully.")
            return
        previous = temp
        temp = temp.next
```

```
    print("Vehicle not found.")

def sort_vehicles():
    global head
    vehicles = []
    temp = head
    while temp:
        vehicles.append(temp.vehicle)
        temp = temp.next
    vehicles.sort()
    print("\nSorted Vehicles:")
    for vehicle in vehicles:
        print(vehicle)

def show_removed():
    print("\nRemoved Vehicles:")
    if not removed:
        print("No removed vehicles.")
    else:
        for vehicle in removed:
            print(vehicle)

def restore_vehicle():
    if not removed:
        print("No vehicle to restore.")
    else:
        vehicle = removed.pop()
        new_node = Node(vehicle)
        global head
        if head is None:
            head = new_node
        else:
            temp = head
            while temp.next:
                temp = temp.next
            temp.next = new_node

        vehicle_data[vehicle] = "Parked"
        print("Last removed vehicle restored:", vehicle)

while True:
    print("\n==============================")
    print("   PARKING MANAGEMENT SYSTEM")
    print("==============================")
```

```python
print("1. Park Vehicle")
print("2. Remove Vehicle")
print("3. Display Vehicles")
print("4. Search Vehicle")
print("5. Sort Vehicles")
print("6. Show Removed Vehicles")
print("7. Restore Last Removed Vehicle")
print("8. Exit")

choice = input("\nEnter your choice: ")
if choice == "1":
    park_vehicle()
elif choice == "2":
    remove_vehicle()
elif choice == "3":
    display_vehicles()
elif choice == "4":
    search_vehicle()
elif choice == "5":
    sort_vehicles()
elif choice == "6":
    show_removed()
elif choice == "7":
    restore_vehicle()
elif choice == "8":
    print("Thank you for using Parking Management System!")
    break
else:
    print("Invalid choice. Please try again.")
```

Output :

```
==============================
   PARKING MANAGEMENT SYSTEM
==============================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 1
Enter vehicle number: MH 04 DR 1212
Vehicle parked successfully.
```

```
==============================
   PARKING MANAGEMENT SYSTEM
==============================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 1
Enter vehicle number: MH 01 WE 3434
Vehicle parked successfully.
```

```
==============================
   PARKING MANAGEMENT SYSTEM
==============================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 1
Enter vehicle number: MH 03 VF 7878
Vehicle parked successfully.
```

```
==============================
   PARKING MANAGEMENT SYSTEM
==============================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 1
Enter vehicle number: MH 02 CT 7850
Vehicle parked successfully.
```

```
==============================
   PARKING MANAGEMENT SYSTEM
==============================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 3

Parked Vehicles:
MH 04 DR 1212
MH 01 WE 3434
MH 03 VF 7878
MH 02 CT 7850
```

```
==============================
   PARKING MANAGEMENT SYSTEM
==============================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 5

Sorted Vehicles:
MH 01 WE 3434
MH 02 CT 7850
MH 03 VF 7878
MH 04 DR 1212
```

```
================================
   PARKING MANAGEMENT SYSTEM
================================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 4
Enter vehicle number to search: MH 02 CT 7850
Vehicle found at position: 4
```

```
================================
   PARKING MANAGEMENT SYSTEM
================================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 2
Enter vehicle number to remove: MH 04 DR 1212
Vehicle removed successfully.
```

```
================================
   PARKING MANAGEMENT SYSTEM
================================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 6

Removed Vehicles:
MH 04 DR 1212
```

```
==============================
   PARKING MANAGEMENT SYSTEM
==============================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 7
Last removed vehicle restored: MH 04 DR 1212
```

```
==============================
   PARKING MANAGEMENT SYSTEM
==============================
1. Park Vehicle
2. Remove Vehicle
3. Display Vehicles
4. Search Vehicle
5. Sort Vehicles
6. Show Removed Vehicles
7. Restore Last Removed Vehicle
8. Exit

Enter your choice: 8
Thank you for using Parking Management System!
```